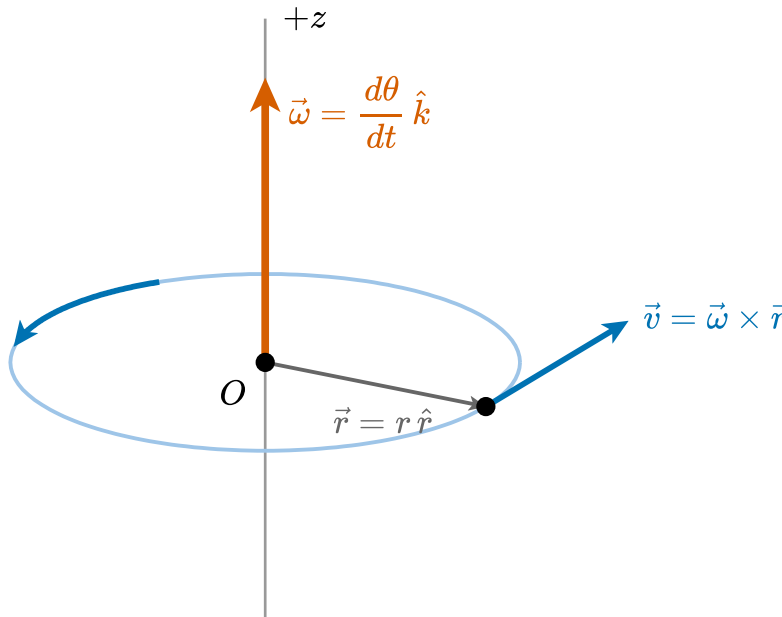




$$\vec{v} = \vec{\omega} \times \vec{r} = \frac{d\theta}{dt} \hat{k} \times r \hat{r} = r \frac{d\theta}{dt} \hat{\theta}$$

$$\|\vec{v}\| = \omega r \sin 90^\circ = \omega r$$